# Geometric Sequences and Series

## Section 1: General Term of a Geometric Sequence

**1** State the formula for the $n$-th term $a_n$ of a geometric sequence with first term $a_1$ and common ratio $r$.

**a.** $a_n = a_1 r^{n-1}$

**b.** $a_n = a_1 r^n$

**c.** $a_n = a_1 + (n-1)r$

**d.** $a_n = \frac{a_1}{r^{n-1}}$

**2** Find the 6th term of the geometric sequence $3, 6, 12, 24, \ldots$

**a.** $96$

**b.** $192$

**c.** $48$

**d.** $128$

**3** In a geometric sequence, $a_2 = 12$ and $a_5 = 324$. Find the common ratio $r$.

**a.** $3$

**b.** $2$

**c.** $4$

**d.** $\sqrt{3}$

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## Section 2: Finite Geometric Series

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**4** State the formula for the sum $S_n$ of the first $n$ terms of a geometric series ($r \neq 1$).

**a.** $S_n = \frac{a_1(1-r^n)}{1-r}$

**b.** $S_n = \frac{a_1}{1-r}$

**c.** $S_n = \frac{n}{2}(a_1 + a_n)$

**d.** $S_n = a_1(1-r^n)$

**5** Find the sum of the first 6 terms of the series $4 + 12 + 36 + \ldots$

**a.** 1456

**b.** 1458

**c.** 486

**d.** 2916

**6** Evaluate $\sum_{k=1}^{5} 2(3)^{k-1}$.

**a.** 242

**b.** 162

**c.** 243

**d.** 486

## Section 3: Infinite Geometric Series

**7** Under what condition does an infinite geometric series $\sum_{n=1}^{\infty} a_1 r^{n-1}$ converge?

**a.** $|r| < 1$

**b.** $r > 1$

**c.** $|r| \leq 1$

**d.** $r > 0$

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**8** Find the sum of the infinite series $12 + 4 + \frac{4}{3} + \frac{4}{9} + \ldots$

**a.** 18

**b.** 16

**c.** 24

**d.** 15

**9** Find the fractional equivalent of the repeating decimal $0.\overline{45} = 0.454545\ldots$ using an infinite geometric series.

**a.** 5/11

**b.** 45/100

**c.** 9/22

**d.** 45/90

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## Section 4: Geometric Word Problems and Annuities

**10** A bouncing ball is dropped from a height of $16$ ft. On each bounce, it rebounds to $3/4$ of its previous height. Find the total vertical distance the ball travels before coming to rest.

**a.** 112 ft

**b.** 64 ft

**c.** 128 ft

**d.** 96 ft

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**11** A company offers a starting salary of $50000$ dollars with a guaranteed $4\%$ annual raise. What is the salary in the 5th year?

**a.** $50000(1.04)^4 \approx 58492.93$ dollars

**b.** $60000$ dollars

**c.** $58000$ dollars

**d.** $50000(1.04)^5$

**12** An ordinary annuity pays $100$ dollars at the end of each year for 10 years at $5\%$ annual interest. Which expression represents its future value?

**a.** $100\left[\frac{(1.05)^{10}-1}{0.05}\right]$

**b.** $100(1.05)^{10}$

**c.** $1000(1.05)$

**d.** $\frac{100}{0.05}$

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